

Compiler Construction Semester Project : **Software Submission : Advice and Instructions**

Submission Deadline : Monday the **27th of October, 7:00h AM in the Morning**, via ClickUp.

- **No extension will be granted** under any circumstances; otherwise our tutors will not have sufficient time to test and to mark the submissions before the end of the semester!
 - **Since a project group has 4 members, there is no "illness" excuse**, because *another group member can still submit the work when one group member is ill.*
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Mode of Submission: ZIP-File-upload, via ClickUp.

The **ZIP-File must contain**:

- Your compiler's **executable run-able software**, ready for immediate deployment and use.
- A short **user-manual (1 page PDF)** which explains to the tutors how they have to operate your software.

IMPORTANT, Note:

- The **tutors do not have the time to inspect your submission's source code!** The tutors work *only* with the Executable! **If your executable "crashes", the tutors will not fix it ! = 0 Points!**
 - The user-manual must also contain an information-declaration, for orienting the tutors about the "type" of your project, whether it is "complete" or "not complete": see below for further details.
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Your software must be able to **"open" and "read" a *.txt** file which will contain an **SPL** Program. Then, **your compiler will be tested** (via numerous test cases) on the basis of such a ***.txt** input file.

TYPES OF PROJECTS **and their marking-value**

Project "Type F": ONLY LEXER

Marking-Value = *maximally* **5 Points**.

INPUT

- SPL Program in a *.txt file

OUTPUT

- **Screen-text-message "Tokens accepted"**, if and only if the given input file did not contain any lexical error
 - **Screen-text-message "Lexical error:""++*some additional helpful information for the user***, in case that the given input file was not free of lexical errors.
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Project "Type E": Complete FRONT-END (including Lexing AND PARSING)

Marking-Value = *maximally* **6 Points**.

INPUT

- SPL Program in a *.txt file

OUTPUT

- All as in Project Type A (of above), PLUS the following additional:
- **Screen-text-message "Syntax accepted"**, if and only if the given input file did not contain any syntax error
- **Screen-text-message "Syntax error:""++*some additional helpful information for the user***, in case that the given input file was not free of syntax errors.

Project "Type D": Front-End PLUS NAME-SCOPE-ANALYSER

Marking-Value = maximally **7 Points**.

INPUT

- SPL Program in a *.txt file

OUTPUT

- All as in Project Type A (of above), PLUS the following additional:
- **Screen-text-message** "Variable Naming and Function Naming accepted", if and only if the given input file did not contain any naming-mistakes
- **Screen-text-message** "Naming error:++some additional helpful information for the user, in case that the given input file was not free of naming-errors in the context of SPL's scoping rules.

Project "Type C": Front-End PLUS complete semantic analysis including TYPE-CHECKER

Marking-Value = maximally **8 Points**.

INPUT

- SPL Program in a *.txt file

OUTPUT

- All as in Project Type A (of above), PLUS the following additional:
- **Screen-text-message** "Types accepted", if and only if the given input file did not contain any type errors
- **Screen-text-message** "Type error:++some additional helpful information for the user, in case that the given input file was not free of type errors in the context of SPL's type-rules.

Project "Type B": Front-End, Semantic Analysis, and Generation of NON-EXECUTABLE INTERMEDIATE CODE

Marking-Value = maximally **9 Points**.

INPUT

- SPL Program in a *.txt file

OUTPUT

- All Screen-text-messages as in Project Type A (of above), PLUS the following additional:
- **If no error messages appeared during the foregoing phases, then** the compiler emits a **HTML-file (*.html)**, which the tutors can nicely inspect via a web-browser: this HTML file shows the generated non-executable Intermediate-Code in a nicely layouted format.

Project "Type A": Front-End, Semantic Analysis, and Generation of EXECUTABLE BASIC CODE

Marking-Value = maximally **10 Points**.

INPUT

- SPL Program in a *.txt file

OUTPUT

- All Screen-text-messages as in Project Type A (of above), PLUS the following additional:
- **If no error messages appeared during the foregoing phases, then** the compiler emits a **text-file (*.txt)**, which contains the generated **executable BASIC program** in the syntax that was described in the most recent task-specification. For assessment, **the tutors will actually run this BASIC program** (via a BASIC-Emulator) in order to see whether the generated code indeed "works" as expected at run-time.

And now:

Best of Luck for the "final preparation" of your Project's Product for Submission and Upload
